Tut 4: Problems for Aug 20

p 70: Find the volume of the Solid obtained by revolving about y axis $g(x) = \frac{\tan^2 x}{x}$, $g(0) = 0$ for $0 \le x \le \frac{\pi}{4}$

p 72: problem 10: Do it with say $m=3$, $n=5$ (Integration by parts)

p 72.1 problem 11

p 72.1 problem 12, 13: Calculate $\int_0^\infty \frac{dx}{1+x^2}$.

Discuss: Notion of Spread of a function over a Closed interval I (Notation $\sigma(I)$)

Give examples like $f(x) = \sin\frac{1}{x}$, $x \neq 0$, $f(0) = 0$

Discuss spread $\sigma([0,c]) = 2$ for every $c > 0$

Show that if $I \subset J$ then $\sigma(I) \le \sigma(J)$

Prove that if (I_n) is a seq of closed intervals all containing a point p and length $I_n \to 0$ if f is continuous at p, then $\sigma(I_n) \to 0$

Tut 5: Aug 27: (1) Do problem on p 70 indicated from Thomas-Finney 9th Ed. p. 393. Using Shell Method. Leave washer method for them to do.

(2) Prove that if $f: [a,b) \to \mathbb{R}$ is continuous, ($a, b \in \mathbb{R}$, $a < b$)
$\exists$ $c \in [a,b]$ s.t. $\int_a^b f(t)\,dt = f(c)(b-a)$

Deduce that $\frac{d}{dt} \int_a^{x} f(t)\,dt = f(x)$.

~~(3) Prove that if $f: [a,b] \to \mathbb{R}$ is continuous~~
~~$\int_a^b f(t)\,dt$ exists~~

Tut 5 Aug 27 (Contd)

Prove that $\left\{ (x,y) \,/\, \frac{x^3}{44} - \frac{y^3}{37} < 1 \right\}$

is an open set.

Is the set $\left\{ (x,y) \,/\, \frac{x^2}{4} + \frac{y^2}{9} < 1 \right\}$ Convex ?

proof ??

~~when f(x,y) =~~

Prove that a polynomial in two variables is a continuous function.

Prove that if $f(x,y)$ is a continuous real valued function $\mathbb{R}^2 \to \mathbb{R}$ then the level set

$\{ (x,y) \in \mathbb{R}^2 \,/\, f(x,y) = c \}$ is closed.

Sketch the level sets of

$f(x,y,z) = x^2 + y^2 - z^2$